$$\mathbf{Q} = \mathcal{R}_M(W_q \mathbf{F}_b^o),$$

$$\mathbf{K} = \mathcal{R}_M(W_k \mathcal{S}_r(\mathbf{F}_b^s)),$$

$$\mathbf{V} = \mathcal{R}_M(W_v \mathcal{S}_r(\mathbf{F}_b^s)).$$